UCI Online Shopping Data Documentation

Overview

This document describes the **UCI Online Shoppers Purchasing Intention Dataset** and the derived datasets stored in the `shopping/data` subfolders.
The original data come from the UCI Machine Learning Repository (Online Shoppers Purchasing Intention Data Set; see UCI entry and related paper, e.g. via Semantic Scholar).

The following datasets are covered here:

- **Raw UCI data**: `data/raw_data/uci_shopping.csv`
- **Preprocessed UCI data**: `data/intermediate_data/01_uci_preprocessed.csv`
- **Uniformly censored UCI data**: `data/intermediate_data/02_uci_uniform_censor.csv`
- **Randomly censored UCI data**: `data/intermediate_data/03_uci_random_censor.csv`

The code that creates these datasets lives in the `shopping/code` folder:

- `00_uci_EDA.Rmd`
- `01_uci_data_preprocessing.R`
- `02_uci_censor_uniform_time.R`
- `03_uci_censor_random_time.R`

Below, we first describe the variables in the **raw** UCI dataset and then explain how each intermediate dataset is constructed, including the key code used.

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1. Raw UCI dataset: `uci_shopping.csv`

File path: `shopping/data/raw_data/uci_shopping.csv`
Source: UCI Machine Learning Repository, Online Shoppers Purchasing Intention Data Set (ID 468).

```
uci_raw <- read_csv(file.path(work_dir, "shopping", "data", "raw_data", "uci_shopping.csv"),
                    show_col_types = FALSE)
glimpse(uci_raw)
```

```
## Rows: 12,330
## Columns: 18
## $ Administrative          <dbl> 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 2~
## $ Administrative_Duration <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 5~
## $ Informational           <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ Informational_Duration  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ ProductRelated          <dbl> 1, 2, 1, 2, 10, 19, 1, 0, 2, 3, 3, 16, 7, 6, 2~
## $ ProductRelated_Duration <dbl> 0.000000, 64.000000, 0.000000, 2.666667, 627.5~
```

```
## $ BounceRates             <dbl> 0.200000000, 0.000000000, 0.200000000, 0.05000~
## $ ExitRates               <dbl> 0.200000000, 0.100000000, 0.200000000, 0.14000~
## $ PageValues              <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ SpecialDay              <dbl> 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.4, 0.0, 0.8, 0~
## $ Month                   <chr> "Feb", "Feb", "Feb", "Feb", "Feb", "Feb", "Feb~
## $ OperatingSystems        <dbl> 1, 2, 4, 3, 3, 2, 2, 1, 2, 2, 1, 1, 1, 2, 3, 1~
## $ Browser                 <dbl> 1, 2, 1, 2, 3, 2, 4, 2, 2, 4, 1, 1, 1, 5, 2, 1~
## $ Region                  <dbl> 1, 1, 9, 2, 1, 1, 3, 1, 2, 1, 3, 4, 1, 1, 3, 9~
## $ TrafficType             <dbl> 1, 2, 3, 4, 4, 3, 3, 5, 3, 2, 3, 3, 3, 3, 3, 3~
## $ VisitorType             <chr> "Returning_Visitor", "Returning_Visitor", "Ret~
## $ Weekend                 <lgl> FALSE, FALSE, FALSE, FALSE, TRUE, FALSE, FALSE~
## $ Revenue                 <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALS~
```

Variable descriptions

The variables below follow the UCI dataset documentation and the explanations in `00_uci_EDA.Rmd`.

Table 1: Raw UCI continuous variables

Variable	Description
Administrative	Number of pages visited that are related to account management (e.g., account settings, order history).
Administrative_Duration	Total time (in seconds) spent on account management–related pages during the session.
Informational	Number of informational pages visited (e.g., pages about the company, contact details, FAQs).
Informational_Duration	Total time (in seconds) spent on informational pages.
ProductRelated	Number of product-related pages visited (e.g., product listings, product detail pages, cart pages).
ProductRelated_Duration	Total time (in seconds) spent on product-related pages.
BounceRates	Average bounce rate of the visited pages in the session; roughly 1 - Engagement Rate, where engagement is based on session length, key events, or multiple page views.
ExitRates	Average exit rate of the visited pages; approximates (Number of Exits / Number of Page Views) x 100 aggregated over pages in the session.
PageValues	Average page value of the visited pages; combines total revenue and goal value per unique page view according to Google Analytics definitions.
SpecialDay	Continuous measure of closeness of the visit date to a special day (e.g., Valentine's Day, Mother's Day); higher values indicate visits closer to the special day.

Table 2: Raw UCI categorical variables

Variable	Description
Month	Month of the visit (e.g., "Feb", "Mar", "May").
OperatingSystems	Encoded identifier of the visitor's operating system.
Browser	Encoded identifier of the visitor's web browser.
Region	Encoded geographic region from which the session was started.
TrafficType	Encoded traffic source indicating how the visitor arrived at the site (e.g., direct, referral, ad campaign).
VisitorType	Visitor type; typically "New_Visitor", "Returning_Visitor", or "Other".
Weekend	Logical indicator; TRUE if the session took place on a weekend, FALSE otherwise.
Revenue	Logical indicator; TRUE if the session ended with a transaction (purchase), FALSE otherwise.

2. Preprocessed UCI dataset: `01_uci_preprocessed.csv`

File path: `shopping/data/intermediate_data/01_uci_preprocessed.csv`
Generating code: `shopping/code/01_uci_data_preprocessing.R`

How `01_uci_preprocessed.csv` is created

Key steps from `01_uci_data_preprocessing.R`:

```
library(tidyverse)
library(here)

work_dir <- here()
raw_data <- read_csv(file.path(work_dir, "shopping", "data", "raw_data", "uci_shopping.csv"))

# Summing up durations
uci_data_summed <- raw_data %>%
  mutate(TotalDuration = Administrative_Duration +
                         Informational_Duration +
                         ProductRelated_Duration) %>%
  select(-c(Administrative_Duration,
            Informational_Duration,
            ProductRelated_Duration)) %>%
  filter(TotalDuration != 0)  # filter out sessions with zero total duration

write.csv(uci_data_summed,
          file.path(work_dir, "shopping", "data", "intermediate_data",
                    "01_uci_preprocessed.csv"),
          row.names = FALSE)
```

Variable changes in `01_uci_preprocessed.csv`

Table 3: Changes from raw UCI data to the preprocessed dataset

Variable	Description
Administrative_Duration. (Removed)	Original per-session time on administrative pages; removed after being summed into TotalDuration.
Informational_Duration (Removed)	Original per-session time on informational pages; removed after being summed into TotalDuration.
ProductRelated_Duration (Removed)	Original per-session time on product-related pages; removed after being summed into TotalDuration.
TotalDuration (New)	Sum of Administrative_Duration + Informational_Duration + ProductRelated_Duration; strictly positive in this dataset.

```
uci_pre <- read_csv(file.path(work_dir, "shopping", "data", "intermediate_data",
                              "01_uci_preprocessed.csv"),
                    show_col_types = FALSE)
glimpse(uci_pre)
```

```
## Rows: 11,610
## Columns: 16
## $ Administrative   <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0,~
```

```
## $ Informational    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ ProductRelated   <dbl> 2, 2, 10, 19, 2, 3, 3, 16, 7, 6, 2, 23, 13, 2, 20, 8,~
## $ BounceRates      <dbl> 0.000000000, 0.050000000, 0.020000000, 0.015789474, 0~
## $ ExitRates        <dbl> 0.100000000, 0.140000000, 0.050000000, 0.024561404, 0~
## $ PageValues       <dbl> 0.00000, 0.00000, 0.00000, 0.00000, 0.00000, 0.00000,~
## $ SpecialDay       <dbl> 0.0, 0.0, 0.0, 0.0, 0.8, 0.4, 0.0, 0.4, 0.0, 0.0, 0.0~
## $ Month            <chr> "Feb", "Feb", "Feb", "Feb", "Feb", "Feb", "Feb", "Feb~
## $ OperatingSystems <dbl> 2, 3, 3, 2, 2, 2, 1, 1, 1, 2, 3, 1, 1, 2, 2, 2, 3, 2,~
## $ Browser          <dbl> 2, 2, 3, 2, 2, 4, 1, 1, 1, 5, 2, 1, 1, 2, 4, 2, 2, 4,~
## $ Region           <dbl> 1, 2, 1, 1, 2, 1, 3, 4, 1, 1, 3, 9, 1, 1, 4, 5, 1, 1,~
## $ TrafficType      <dbl> 2, 4, 4, 3, 3, 2, 3, 3, 3, 3, 3, 3, 4, 3, 4, 1, 5, 3,~
## $ VisitorType      <chr> "Returning_Visitor", "Returning_Visitor", "Returning_~
## $ Weekend          <lgl> FALSE, FALSE, TRUE, FALSE, FALSE, FALSE, FALSE, FALSE~
## $ Revenue          <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALS~
## $ TotalDuration    <dbl> 64.000000, 2.666667, 627.500000, 154.216667, 37.00000~
```

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3. Uniformly censored UCI dataset: `02_uci_uniform_censor.csv`

File path: `shopping/data/intermediate_data/02_uci_uniform_censor.csv`
Generating code: `shopping/code/02_uci_censor_uniform_time.R`

How `02_uci_uniform_censor.csv` is created

Key steps from `02_uci_censor_uniform_time.R`:

```r
library(tidyverse)
library(here)

work_dir <- here()
uci_data <- read_csv(file.path(work_dir, "shopping", "data", "intermediate_data",
                               "01_uci_preprocessed.csv"))

# Introducing censorship with a single uniform time threshold
set.seed(123)
time_threshold <- runif(1, min = 10, max = 1800)  # between 10 seconds and 30 minutes

uci_transformed <- uci_data %>%
  mutate(
    TimeThreshold = time_threshold,
    CensorTime    = ifelse(TotalDuration >= time_threshold,
                           time_threshold,
                           TotalDuration),
    Censor        = ifelse(TotalDuration >= time_threshold, 1, 0)
  )

write.csv(uci_transformed,
          file.path(work_dir, "shopping", "data", "intermediate_data",
                    "02_uci_uniform_censor.csv"),
          row.names = FALSE)
```

New variables in `02_uci_uniform_censor.csv`

Starting from `01_uci_preprocessed.csv`, this dataset adds three new variables:

- **`TimeThreshold`**:
  - A scalar threshold (same value for all observations).
  - Drawn from a continuous uniform distribution on $[10, 1800]$ seconds.
  - Represents a global censoring time.
- **`CensorTime`**:
  - The observed time after applying censoring:

$$\text{CensorTime} = \begin{cases} \text{TotalDuration}, & \text{if TotalDuration} < \text{TimeThreshold}, \\ \text{TimeThreshold}, & \text{if TotalDuration} \geq \text{TimeThreshold}. \end{cases}$$

- **`Censor`**:
  - Indicator of whether an observation is censored at `TimeThreshold`:
    - `1` if `TotalDuration >= TimeThreshold` (censored).
    - `0` otherwise (fully observed).

Table 4: Additional variables in the uniformly censored UCI dataset

Variable	Description
TimeThreshold	Scalar censoring threshold (same value for all observations), drawn from a continuous uniform distribution on [10, 1800] seconds; represents a global censoring time.
CensorTime	Observed time after applying censoring: equals TotalDuration if TotalDuration < TimeThreshold, otherwise equals TimeThreshold.
Censor	Censoring indicator: 1 if TotalDuration >= TimeThreshold (censored at the threshold), 0 otherwise.

All other variables from `01_uci_preprocessed.csv` are preserved unchanged.

```r
uci_uniform <- read_csv(file.path(work_dir, "shopping", "data", "intermediate_data",
                                  "02_uci_uniform_censor.csv"),
                        show_col_types = FALSE)
glimpse(uci_uniform)
```

```
## Rows: 11,610
## Columns: 19
## $ Administrative   <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0,~
## $ Informational    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ ProductRelated   <dbl> 2, 2, 10, 19, 2, 3, 3, 16, 7, 6, 2, 23, 13, 2, 20, 8,~
## $ BounceRates      <dbl> 0.000000000, 0.050000000, 0.020000000, 0.015789474, 0~
## $ ExitRates        <dbl> 0.100000000, 0.140000000, 0.050000000, 0.024561404, 0~
## $ PageValues       <dbl> 0.00000, 0.00000, 0.00000, 0.00000, 0.00000, 0.00000,~
## $ SpecialDay       <dbl> 0.0, 0.0, 0.0, 0.0, 0.8, 0.4, 0.0, 0.4, 0.0, 0.0, 0.0~
## $ Month            <chr> "Feb", "Feb", "Feb", "Feb", "Feb", "Feb", "Feb", "Feb~
## $ OperatingSystems <dbl> 2, 3, 3, 2, 2, 2, 1, 1, 1, 2, 3, 1, 1, 2, 2, 2, 3, 2,~
## $ Browser          <dbl> 2, 2, 3, 2, 2, 4, 1, 1, 1, 5, 2, 1, 1, 2, 4, 2, 2, 4,~
## $ Region           <dbl> 1, 2, 1, 1, 2, 1, 3, 4, 1, 1, 3, 9, 1, 1, 4, 5, 1, 1,~
```

```
## $ TrafficType      <dbl> 2, 4, 4, 3, 3, 2, 3, 3, 3, 3, 3, 3, 4, 3, 4, 1, 5, 3,~
## $ VisitorType      <chr> "Returning_Visitor", "Returning_Visitor", "Returning_~
## $ Weekend          <lgl> FALSE, FALSE, TRUE, FALSE, FALSE, FALSE, FALSE, FALSE~
## $ Revenue          <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALS~
## $ TotalDuration    <dbl> 64.000000, 2.666667, 627.500000, 154.216667, 37.00000~
## $ TimeThreshold    <dbl> 524.7638, 524.7638, 524.7638, 524.7638, 524.7638, 524~
## $ CensorTime       <dbl> 64.000000, 2.666667, 524.763761, 154.216667, 37.00000~
## $ Censor           <dbl> 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0,~
```

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4. Randomly censored UCI dataset: `03_uci_random_censor.csv`

File path: `shopping/data/intermediate_data/03_uci_random_censor.csv`
Generating code: `shopping/code/03_uci_censor_random_time.R`

How `03_uci_random_censor.csv` is created

Key steps from `03_uci_censor_random_time.R`:

```r
library(tidyverse)
library(here)

work_dir <- here()
uci_data <- read_csv(file.path(work_dir, "shopping", "data", "intermediate_data",
                               "01_uci_preprocessed.csv"))

# Introducing censorship with row-specific random thresholds
set.seed(123)
time_threshold <- runif(nrow(uci_data), min = 10, max = 1800)  # one threshold per row

uci_transformed <- uci_data %>%
  mutate(
    TimeThreshold = time_threshold,
    CensorTime    = ifelse(TotalDuration >= time_threshold,
                           time_threshold,
                           TotalDuration),
    Censor        = ifelse(TotalDuration >= time_threshold, 1, 0)
  )

write.csv(uci_transformed,
          file.path(work_dir, "shopping", "data", "intermediate_data",
                    "03_uci_random_censor.csv"),
          row.names = FALSE)
```

New variables in `03_uci_random_censor.csv`

Relative to `01_uci_preprocessed.csv`, the same three censoring variables are added as in the uniform case, but with **observation-specific** thresholds:

- **`TimeThreshold`**:

  - A numeric vector of length n, with one value per observation.
  - Each value is drawn independently from $[10, 1800]$ seconds.

- **`CensorTime`**:
  - Row-wise censored time:

$$\text{CensorTime}_i = \begin{cases} \text{TotalDuration}_i, & \text{if TotalDuration}_i < \text{TimeThreshold}_i, \\ \text{TimeThreshold}_i, & \text{if TotalDuration}_i \geq \text{TimeThreshold}_i. \end{cases}$$

- **`Censor`**:
  - Row-wise censoring indicator:
    * `1` if `TotalDuration_i >= TimeThreshold_i` (censored).
    * `0` otherwise.

Table 5: Additional variables in the randomly censored UCI dataset

Variable	Description
TimeThreshold	Numeric vector of length n (one value per observation); each value is drawn independently from a uniform distribution on [10, 1800] seconds.
CensorTime	Row-wise censored time: equals TotalDuration_i if TotalDuration_i < TimeThreshold_i, otherwise equals TimeThreshold_i.
Censor	Row-wise censoring indicator: 1 if TotalDuration_i >= TimeThreshold_i (censored), 0 otherwise.

All other variables (from `01_uci_preprocessed.csv`) remain unchanged.

```r
uci_random <- read_csv(file.path(work_dir, "shopping", "data", "intermediate_data",
                                 "03_uci_random_censor.csv"),
                       show_col_types = FALSE)
glimpse(uci_random)
```

```
## Rows: 11,610
## Columns: 19
## $ Administrative   <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0,~
## $ Informational    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ ProductRelated   <dbl> 2, 2, 10, 19, 2, 3, 3, 16, 7, 6, 2, 23, 13, 2, 20, 8,~
## $ BounceRates      <dbl> 0.000000000, 0.050000000, 0.020000000, 0.015789474, 0~
## $ ExitRates        <dbl> 0.100000000, 0.140000000, 0.050000000, 0.024561404, 0~
## $ PageValues       <dbl> 0.00000, 0.00000, 0.00000, 0.00000, 0.00000, 0.00000,~
## $ SpecialDay       <dbl> 0.0, 0.0, 0.0, 0.0, 0.8, 0.4, 0.0, 0.4, 0.0, 0.0, 0.0~
## $ Month            <chr> "Feb", "Feb", "Feb", "Feb", "Feb", "Feb", "Feb", "Feb~
## $ OperatingSystems <dbl> 2, 3, 3, 2, 2, 2, 1, 1, 1, 2, 3, 1, 1, 2, 2, 2, 3, 2,~
## $ Browser          <dbl> 2, 2, 3, 2, 2, 4, 1, 1, 1, 5, 2, 1, 1, 2, 4, 2, 2, 4,~
## $ Region           <dbl> 1, 2, 1, 1, 2, 1, 3, 4, 1, 1, 3, 9, 1, 1, 4, 5, 1, 1,~
## $ TrafficType      <dbl> 2, 4, 4, 3, 3, 2, 3, 3, 3, 3, 3, 3, 4, 3, 4, 1, 5, 3,~
## $ VisitorType      <chr> "Returning_Visitor", "Returning_Visitor", "Returning_~
## $ Weekend          <lgl> FALSE, FALSE, TRUE, FALSE, FALSE, FALSE, FALSE, FALSE~
## $ Revenue          <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALS~
## $ TotalDuration    <dbl> 64.000000, 2.666667, 627.500000, 154.216667, 37.00000~
## $ TimeThreshold    <dbl> 524.76376, 1421.06619, 742.06869, 1590.60115, 1693.43~
## $ CensorTime       <dbl> 64.000000, 2.666667, 627.500000, 154.216667, 37.00000~
## $ Censor           <dbl> 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0,~
```

5. Relationship between datasets and code files

- **Exploratory data analysis and variable explanations** for the raw UCI dataset are in `shopping/code/00_uci_EDA.Rmd`.

- **Creation of `01_uci_preprocessed.csv`** is handled by `shopping/code/01_uci_data_preprocessing.R`, which:
  - Reads `uci_shopping.csv`.
  - Sums duration variables into `TotalDuration`.
  - Drops zero–total-duration sessions.
  - Writes the result to `01_uci_preprocessed.csv`.

- **Creation of `02_uci_uniform_censor.csv`** is handled by `shopping/code/02_uci_censor_uniform_time.R`, which:
  - Reads `01_uci_preprocessed.csv`.
  - Draws a **single** `TimeThreshold` from $[10, 1800]$ seconds.
  - Creates `CensorTime` and `Censor` based on that threshold.
  - Writes the result to `02_uci_uniform_censor.csv`.

- **Creation of `03_uci_random_censor.csv`** is handled by `shopping/code/03_uci_censor_random_time.R`, which:
  - Reads `01_uci_preprocessed.csv`.
  - Draws **row-specific** `TimeThreshold` values from $[10, 1800]$ seconds.
  - Creates `CensorTime` and `Censor` for each row.
  - Writes the result to `03_uci_random_censor.csv`.

Together, these files document the full pipeline from the original UCI Online Shoppers Purchasing Intention data to the censored datasets used for subsequent analysis.